

TrES-2b

R-1763

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_150905 · created 2026-08-02T21:12:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457270.72844 +/- 0.0004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.043
Transit depth (Rp/Rs)^2	2.19 +/- 1.28 [%]
Orbital Inclination (inc)	88.37 +/- 2.85
Airmass coefficient 1 (a1)	0.96 +/- 0.6
Airmass coefficient 2 (a2)	0.02 +/- 0.93
Scatter in the residuals of the lightcurve fit is	64.82 %
Best Comparison Star	None
Optimal Aperture	1.9
Optimal Annulus	3.35
Transit Duration (day)	0.103 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.043 vs 0.1254 (deviation 0.47σ)
Duration fit vs published	0.103 d vs 0.0746 d (deviation 1.33σ)
Mid-time O–C	0.6 min
Depth SNR	3.1
Residual scatter	64.82 %

Light curve

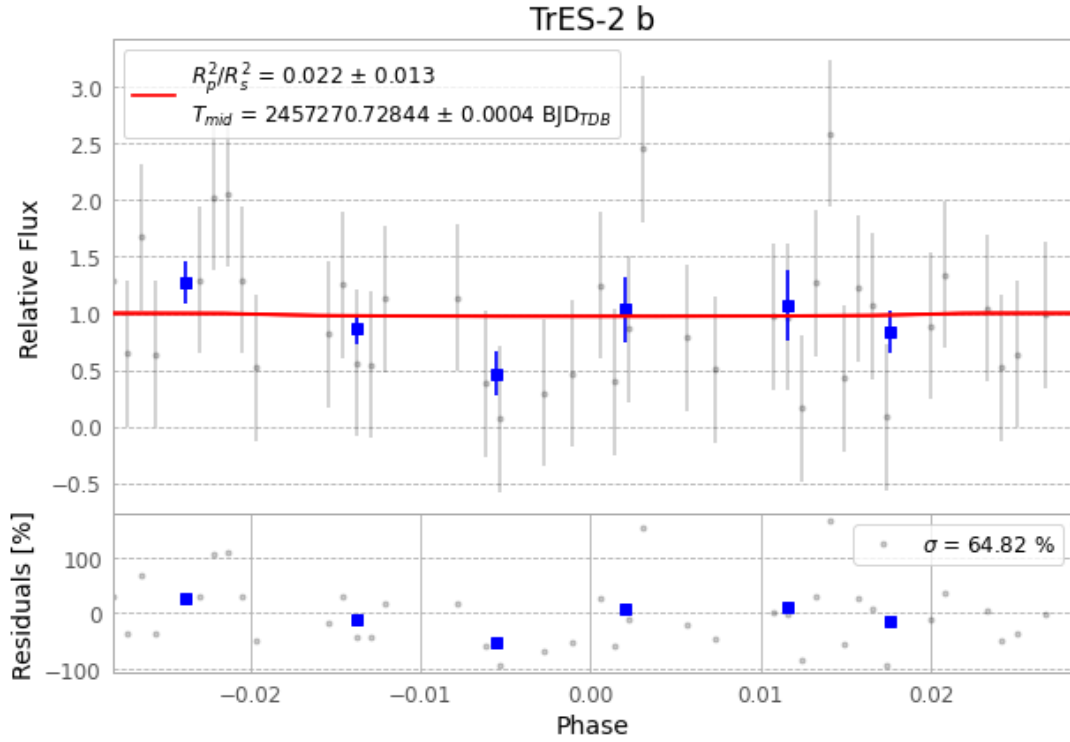


Figure 1 — FinalLightCurve_TrES-2 b_2015-09-04.png (SHA-256 04235a3b1bd90656...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [62, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dbd25be91c18e934...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), TRES-2150905034514.FITS → TRES-2150905070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-150905.log (90e8ea809f14...), FinalLightCurve_TrES-2 b_2015-09-04.png (04235a3b1bd9...), FinalLightCurve_TrES-2 b_2015-09-04.csv (40d45af842d7...)

Compute resources

Wall time 260.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 21:12Z.